Machine Learning 741 – Assignment 4

Option 4: An Ensemble-based Breast Cancer Predictive Model

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***Abstract*—This report compares a homogeneous ensemble and a heterogeneous ensemble for breast tumour classification. The homogeneous ensemble is a Random Forest (RF). The heterogeneous ensemble combines Logistic Regression (LR), k-Nearest Neighbours (kNN), Decision Tree (DT), Gradient Boosting (GB), and Support Vector Machine (SVM) with soft voting. The goal is to identify the approach that attains stronger and more stable performance on the breast cancer dataset.**

Preprocessing of the dataset removes identifiers and constant features, addresses missing values with mean imputation, clamps outliers using interquartile range bounds, and evaluates the effect of class imbalance. Hyper-parameters are tuned with grid search. Evaluation uses cross-validation and multiple metrics: accuracy, weighted $F1$, Matthews Correlation Coefficient (MCC) scores, and receiver operating characteristic (ROC). For the RF, the out-of-bag (OOB) estimate provides an internal validation signal. Reproducibility is ensured by fixed random seeds and clearly specified folds.

Across the primary metrics, the RF attains the strongest overall performance on the original dataset. The heterogeneous soft-voting ensemble is competitive but does not surpass the RF. Balancing the class distribution does not improve performance and, in several cases, reduces mean accuracy, weighted $F1$, and MCC scores, which suggests limited benefit from synthetic sampling for this task. The results indicate that a well tuned RF provides a strong and robust baseline for this classification problem, while heterogeneous ensembles performs slightly worse, but still gives good results.

I. Introduction

Accurate classification of breast tumours supports clinical decision making and risk stratification. Ensemble learning improves predictive performance by combining multiple models. This report compares a homogeneous ensemble to a heterogeneous ensemble on the breast cancer dataset, with the aim of identifying the stronger and more stable approach.

The homogeneous ensemble is a Random Forest (RF). The heterogeneous ensemble combines Logistic Regression (LR), k-Nearest Neighbours (kNN), Decision Tree (DT), Gradient Boosting (GB), and Support Vector Machine (SVM) via soft voting, where class probabilities are averaged. Hyper-parameters are tuned via a grid search method. Evaluation uses cross-validation and multiple metrics which include accuracy, weighted $F1$, Matthews Correlation Coefficient (MCC) scores, receiver operating characteristic (ROC).

Preprocessing removes identifiers and constant features, addresses missing values and outliers, and examines the effect of class imbalance. Results show that the RF attains the strongest overall performance, while the heterogeneous ensemble is competitive but does not surpass the RF.

Section II provides background on ensembles and voting. Section III details implementation and preprocessing. Section IV presents the empirical procedure and metrics. Section V reports and discusses results. Section VI concludes with key findings and future work.

II. Background

This section discusses important background concepts that will give an understanding of the dataset used, what are homogeneous ensembles, what are heterogenous ensembles and the metrics used to evaluate the performance of the homogeneous and heterogeneous ensembles, respectively.

A. The Breast Cancer Dataset

The breast cancer dataset is a dataset about breast cancer tumours. The target feature is *diagnosis* which indicates if a tumour is malignant (M) or benign (B). The dataset contains the following descriptive features: *id*, *diagnosis*, *radius_mean*, *texture_mean*, *perimeter_mean*, *area_mean*, *smoothness_mean*, *compactness_mean*, *concavity_mean*, *concave_points mean*, *symmetry_mean*, *fractal_dimension_mean*, *radius_se*, *texture_se*, *perimeter_se*, *area_se*, *smoothness_se*, *compactness_se*, *concavity_se*, *concave_points_se*, *symmetry_se*, *fractal_dimension_se*, *radius_worst*, *texture_worst*, *perimeter_worst*, *area_worst*, *smoothness_worst*, *compactness_worst*, *concavity_worst*, *concave_points_worst*, *symmetry_worst*, *fractal_dimension_worst*, *gender* and *Bratio*.

Within this dataset there were missing value present, outliers, unique identifiers, constant features and a skewed class distribution. In the **Methodology** section, each data quality issue was addressed and a method was implemented to fix the data quality issues.

B. Machine Learning Algorithms

The objective of this study was to evaluate and compare the performance of homogeneous and heterogeneous ensemble methods using a breast cancer dataset. Ensemble methods operate on the principle that collective predictions from multiple models yield more robust and accurate results than a single model. This approach is derived from the "wisdom-of-the-crowd" phenomenon, wherein the decisions of a large group consistently outperform those of individual experts. In machine learning, a single algorithm trained on a dataset may identify a specific set of patterns, resulting in a certain level of accuracy. In contrast, a properly tuned ensemble combines multiple algorithms, each capable of capturing different patterns in the data. The union of these models aims to achieve a performance that meets or, more desirably, exceeds that of the best individual model.

For the homogeneous ensemble, the Random-Forest (RF) algorithm was implemented and for the heterogeneous ensemble, the Logistic-Regression (LR), k-Nearest Neighbour (kNN), Decision Trees (DT), Gradient Boosting (GB) and Support Vector Machine (SVM) was used. The following two sub-sub-subsections present the reasoning behind why the the algorithms for the homogeneous and heterogenous ensembles were utilised, respectively.

1) Homogeneous Ensemble: A homogeneous ensemble employs a single type of machine learning algorithm for all of its constituent models. An example is the RF algorithm, which utilises a bagging methodology with DT as its base learners [1]. In bagging, multiple data subsets are created by randomly sampling the training set with replacement, allowing individual data points to be selected more than once [1]. Each of these subsets is then used to train a separate “weak” model independently. The idea of a “weak” model does not mean it is a bad model, but it would not be as optimised as a “strong” model. These type of models (“weak” models) should perform better than random guessing, they are light (computationally inexpensive) and individually trained. An example of a “weak” model is a DT with a small depth.

In this study, a RT was employed for use in the homogeneous ensemble. Figure 1 shows a graphical representation of a RT,

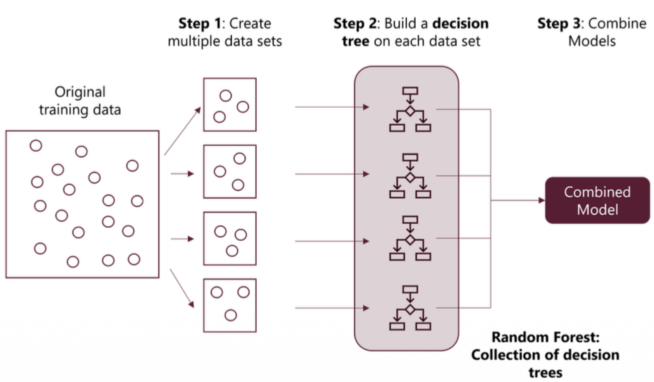

Fig. 1. Graphical illustration of a RT.

Figure 1 shows the process behind employing a RF. Firstly, the original training data is split into multiple datasets which the Dt will be trained and tuned on. Secondly, build a DT on each of the new dataset and finally, combine the models to make a RF.

The RF algorithm was selected for the homogeneous ensemble for the following reasons:

- DT work well with any type of data.
- Robust to outliers and noise.
- Robust to missing values.
- It is the canonical example of a homogeneous ensemble.

Due to its ability of robustness against specific data quality issues and also its implementation is very simply, it was decided to use the RF for the homogeneous ensemble.

2) Heterogenous Ensemble: A heterogeneous ensemble comprises multiple machine learning algorithms, whose individual predictions are aggregated to produce a final output. The following list is some reasons as to why these algorithms were employed:

- **LR:** It is a linear model which is fast, gives results that are interpretable, and captures global trends.
- **kNN:** Depending on value of k will either be sensitive or robust to outliers. But because these outliers were addressed, kNN was utilised without the worry of affects of outliers.
- **DT:** Robust to outliers and noise, skewed class distributions. Furthermore, was used in the hyper-parameter tuning for RF hence, was also included in the heterogeneous ensemble.
- **GB:** Subsequent models try to reduce the errors of previous models, by building a new model on the actual errors or residuals of the previous model, and not the actual target values.
- **SVM:** Adds robustness to high-dimensional data and captures complex non-linear boundaries [2].

By combining these diverse classifiers using soft voting, the ensemble benefits from the strengths of each individual learner, which could lead to improved robustness, improved stability, and better overall predictive performance.

The following pseudo-code are for each algorithm that was used in the heterogeneous ensemble,

Algorithm 1 Logistic Regression Algorithm

Input: Training data (x_i, y_i) for $i = 1, \ldots, N$, learning rate η, number of iterations T
Initialize weights $w \leftarrow 0$, bias $b \leftarrow 0$
for $t = 1$ to T **do**
  for each training example (x_i, y_i) **do**
    $\hat{y}_i \leftarrow \sigma(w^\top x_i + b)$ {Compute predicted probability}
    $w \leftarrow w - \eta(\hat{y}_i - y_i)x_i$ {Update weights}
    $b \leftarrow b - \eta(\hat{y}_i - y_i)$ {Update bias}
  end for
end for
Output: Final model parameters w, b

Algorithm 2 kNN Algorithm

function KNN(D, x, k)
for all $x_i \in D$ **do**
  $\mathbf{d}$ = DISTANCE(x_i, x)
end for
SORT($\mathbf{d}$)
S = set of k patterns in D closest to x
Return class as majority class in S
end function

Algorithm 3 Decision Tree Algorithm

function INDUCETREE(D)
if $|D| = 0$ **then**
  Return leaf with default class ▷ D is empty
end if
if $|D| > 0$ **and** all $x \in D$ have same class y_m **then**
  Return leaf with class label y_m, containing D ▷ D is homogeneous
end if
Select a test based on a single descriptive feature ▷ D not homogeneous
Split D into $D_1, D_2, \ldots, D_O$, where O is number of outcomes
for $o = 1$ **to** O **do**
  INDUCETREE(D_o)
end for
end function

Algorithm 4 Gradient Boosting Classifier Algorithm [3]

Input: Input features $x^{(1:n)} \in \mathbb{R}^{n \times d}$ and targets $y^{(1:n)} \in \{0,1\}^n$
Input: Number of Trees M
Input: Learning rate η
Output: Fitted trees $T_{1:M}$
// Initialization
for $i = 1$ to n **do**
  $F_0^{(i)} \leftarrow 0$
  $p_0^{(i)} \leftarrow \sigma(F_0^{(i)}) = 0.5$
end for
// Training for M iterations
for $m = 1$ to M **do**
  for $i = 1$ to n **do**
    Compute residual: $r_m^{(i)} \leftarrow y^{(i)} - p_{m-1}^{(i)}$
  end for
  Train a regression tree T_m to fit the dataset $\{(x^{(i)}, r_m^{(i)})\}_{i=1,\ldots,n}$
  // Update terminal node values
  Let $\Omega_{m,j}$ be the set of instance indices in terminal node j of T_m
  for each terminal node $j \in T_m$ **do**
    Set the output of node j in T_m: $\gamma_j \leftarrow \dfrac{\sum_{k \in \Omega_{m,j}} r_m^{(k)}}{\sum_{k \in \Omega_{m,j}} p_{m-1}^{(k)}(1 - p_{m-1}^{(k)})}$
    for $k \in \Omega_{m,j}$ **do**
      $F_m^{(k)} \leftarrow F_{m-1}^{(k)} + \eta\gamma_j$
      $p_m^{(k)} \leftarrow \sigma(F_m^{(k)})$
    end for
  end for
end for

Algorithm 5 Support Vector Machine Algorithm

Initialize weight vector w and bias b
Set learning rate η and regularization parameter C
$t = 0$
while SVM has not converged **do**
  Select training pattern (x_i, y_i)
  if $y_i(w^\top x_i + b) < 1$ **then**
    $w \leftarrow w + \eta(y_i x_i - 2Cw)$
    $b \leftarrow b + \eta y_i$
  else
    $w \leftarrow w - \eta(2Cw)$
  end if
  $t \leftarrow t + 1$
end while

C. Voting

Soft voting, is an ensemble method that aggregates the class probability vectors from each base model. The final prediction is determined by computing the weighted average of these probabilities across all models and subsequently selecting the class with the highest resultant score. For the heterogenous ensemble a weighted soft voting approach was utilised.

III. Methodology

This section outlines the methodological framework of this study, including the procedures for data quality issues and the implementation of the machine learning algorithms.

A. Programming Environment and Libraries

All data preprocessing and machine learning implementation were conducted in *Python*, utilizing the VS Code environment for development, experimentation, and evaluation of model performance.

This study relied on specific libraries available for *Python*, these include:

- **Numpy:** Package for scientific computing. Was used for missing value correction and outlier correction.
- **Pandas:** Package for easy-to-use data structures and data analysis tools. Was used for reading in breast cancer dataset and general checking of data.
- **Matplotlib:** Package for creating visualisations. Any plotting was done using this package.
- **Scikit-learn:** This package is a Python machine learning library that provides simple and efficient tools for data analysis and modelling.
- **Imbalanced-learn:** This package is a Python library that specialises in handling imbalanced datasets situations where one class has significantly more samples than another. It provides techniques for resampling data.

B. Implementation

This subsection describes the methods behind how the data quality issues were addressed and how the homogeneous ensemble and heterogenous ensembles were implemented.

1) Data Preprocessing: As previously mentioned, this dataset had some data quality issues. These included missing value, outliers, unique identifiers, constant features, and skewed class distributions. Using **Pandas** we addressed each data quality issue in the following way:

1) Read in *csv* file and print out file.
2) Split file into descriptive features and target feature by dropping target feature.
3) Drop unique identifier (*id* feature) and drop constant feature (*gender* feature).
4) Count missing values per feature (checks if there ia any). Calculate mean of valid instances in feature using **numpy.mean()** then, replace missing value with mean value.
5) Calculate interquartile range per feature. Count amount of outliers and clip outliers to lower and upper bounds.
6) The class imbalance was addressed only after the initial models were trained and evaluated on the unmodified dataset.

The steps shown completes the data preprocessing implemented in this study.

2) Justification for Data Preprocessing: The RF algorithm is robust to all the data quality issues shown previously, however, algorithms used in the heterogeneous ensemble are not robust to missing values, outliers or skewed class distributions. For example, the kNN algorithm is not always robust to outliers (depends on the value of k). Hence, to keep the training and testing fair for both the homogenous ensemble and the heterogeneous ensemble, the data quality issues were addressed.

The data quality issues were addressed as follows. Given the low prevalence of missing values and outliers across descriptive features, mean imputation was applied to preserve the underlying distribution of the data. Outliers were clamped to the lower and upper bounds of the interquartile range to maintain feature distribution without significantly impacting ensemble performance. Unique identifiers and constant features were removed, as they lack predictive power and their inclusion could adversely affect results and increase training and testing times of the ensembles.

Finally, to evaluate the impact of class imbalance, both ensembles were trained and tested on both the original dataset and a version with a rectified class distribution. This comparative analysis was conducted because certain algorithms, such as RF and DT, are known to be robust to such skews and ultimately, the nature of the data set is a skewed class distribution.

Figure 2 is histograms of all the features to illustrate the class distributions,

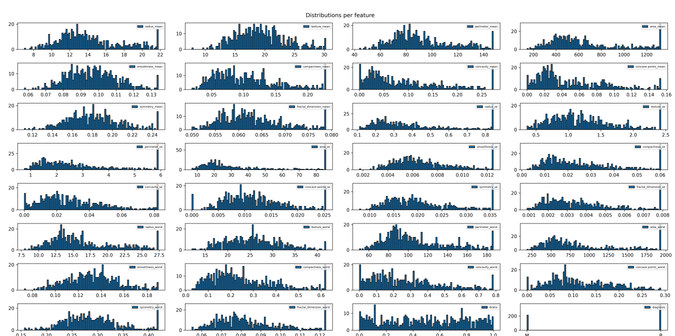

Fig. 2. Distributions of all features in the dataset.

Most features exhibit skewness, though it is not pronounced.

IV. Empirical Procedure

This section outlines the empirical methodology employed to evaluate the performance of the homogeneous and heterogeneous ensemble models on the breast cancer dataset. It details the hyper-parameter tuning strategy, the ensemble configurations, the cross-validation procedure, and the performance metrics used to evaluate performance.

A. Hyper-parameter Tuning

Hyper-parameter tuning was performed for both the homogeneous and heterogeneous ensembles. Similar approaches were used for both ensembles.

For the homogeneous ensemble, hyper-parameter tuning was first performed on a DT to see what are the best hyper-parameter values. Using *GridSearchCV* from *scikit-learn*, a grid search was performed over the hyper-parameter values, *criterion*, *max_depth*, *min_samples_split* and *min_samples_leaf*. A grid search was performed on a DT first because RF is a homogeneous ensemble of DT, hence, finding the best hyper-parameter values for a DT should translate into the best hyper-parameter values for the RF. All hyper-parameters were placed in a Python dictionary, whilst the *scoring* parameter for the grid search was set to 'accuracy'. Furthermore, the *cv* parameter for the grid search function was set to 5. This results in 5-fold cross validation across the different hyper-parameter values. *cv* was set to 5 because firstly, it give reliable results and secondly, would produce results quickly. Once the grid search was completed, the best parameters with its best cross-validation accuracy was displayed.

After this, further hyper-parameter tuning was performed on the RF. Using *scikit-learn*'s *RandomForestClassifier* function, a RF was implemented. The RF has two additional parameters not present in the DT, those where *n_estimators* and *oob_score*. The first being the amount of estimators present in the RF and the second being, when set to True would enable the user to acquire the out-of-bag score (refer to the next sub-section for an explanation). For the RF, only the *n_estimator* was tuned whilst the other hyper-parameters were set to the best hyper-parameter values from the grid search.

A methodology similar to the homogeneous ensemble was employed for the heterogeneous ensemble, wherein each machine learning algorithm underwent hyper-parameter tuning. A grid search was utilised for all machine learning algorithms within the heterogeneous ensemble. 5-fold cross validation was implemented for hyper-parameter tuning and the best parameter values for each machine algorithm was then used to build the heterogeneous ensemble. Thereafter, 10-fold cross validation was performed on the dataset whilst the heterogeneous ensemble was employed to make predictions.

To see the best hyper-parameter values found by *GridSearchCV* refer to Section 5, Research Results.

B. Performance Metrics

This sub-section describes the evaluation metrics, including accuracy, F1-score, Matthews' correlation coefficient (MCC), confusion matrices, out-of-bag score and Receiver operating characteristic curve (ROC) used to assess the performance of the homogeneous and heterogeneous ensembles.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \in [0, 1] \tag{1}$$

Equation 1 describes the formula used to calculate the accuracy score. In *scikit-learn* the *accuracy_score()* function is used to acquire the accuracy between the target feature and the predictions from the model. TP, TN, FP, FN are the true positive, true negative, false positive and false negative values, respectively. True positives are correctly predicted positive outcome, true negatives are correctly predicted negative outcome, false positives are negative instance predicted to be positive and false negatives are positive instance predicted to be negative. Accuracy was chosen as a performance metric because is measures how often the model's predictions are correct overall. It is useful as a general performance indicator.

$F1$ score was used as a performance metric. Equation (2) and (3) illustrates the formula to calculate the weighted F1 score.

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} \times \text{recall}} \tag{2}$$

$$\text{Weighted } F1 = \sum_{i=1}^{N} w_i \times F1 \tag{3}$$

The precision term in equation (2) focuses on the quality of the models positive predictions and shows how many instances are predicted as positive are actually positive. Whereas, the recall term measures how well the model identifies all actual positive cases and shows the proportion of true positives detected out of all the actual positive instances. The weighted $F1$ score accounts for class imbalance by combining precision and recall while weighting each class by its support. This ensures the model's performance is not dominated by the majority class.

Matthews' correlation coefficient is calculated as the following:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \in [-1, 1] \tag{4}$$

The MCC measures the correlation between the real and the predicted values of the instances and is robust to class imbalances.

The accuracy, weighted $F1$ and MCC scores were implemented so that even though there is a class imbalance present, the scores would not have any bias to them, whereas the accuracy score is to give a general baseline to the performance of the ensemble.

Furthermore, confusion matrices and receiver operating characteristic curve were used to evaluate performance. These metrics were only used after hyper-parameter tuning was done, the best values were picked, and testing was done on ensembles. The confusion matrix simply represents the prediction summary in a table form, which can help identify where the model gets confused and which classes are frequently misclassified. The ROC curve is used because it evaluates model performance across all classification thresholds, showing the trade-off between true positives and false positives. Figures 3 and 4 give an example of a confusion matrix and ROC curve, respectively:

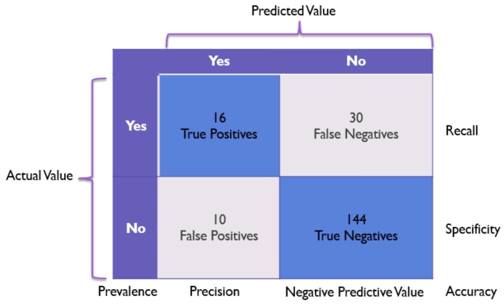

Fig. 3. Example of a confusion matrix.

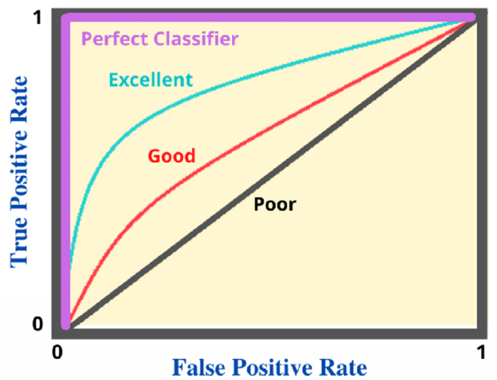

Fig. 4. Example of ROC curve.

The last performance metric used, but only for the homogeneous ensemble was the out-of-bag score (OOB), this is because the OOB score is only applicable to bagging based methods. The heterogeneous ensemble does not use any bagging based methods. The OOB score provides an internal validation estimate for the RF by evaluating each tree on samples not used during its training. It offers a built in, unbiased performance estimate without needing a separate validation set, making it efficient and reliable for assessing generalisation in this study.

C. Cross Validation

Hyper-parameter tuning was conducted using k-fold cross-validation. An initial search was performed with 5-fold cross validation to efficiently identify promising hyper-parameter ranges. For the final evaluation of both homogeneous and heterogeneous ensembles, a more rigorous 10-fold cross validation was employed to ensure reliable performance estimates. By dividing the data into ten subsets, the model is trained on nine folds and tested on the remaining one, repeating this process ten times. This approach reduces variance in the results, ensures that every sample is used for both training and validation, and gives a more robust assessment of generalisation compared to a single train-test split.

D. Reproducibility

To ensure consistent and fair comparisons between the homogeneous and heterogeneous ensembles, a fixed random seed was applied throughout all experiments where we set the parameter *random_state* equal to 42. This ensures that the hyper-parameter tuning, training and evaluation of each ensemble is under identical conditions, allowing performance differences to be attributed solely to the model architectures and ensemble methods rather than random variation.

V. Research Results

This section displays and discusses the performance of the homogeneous and heterogeneous ensembles using the performance metrics described in the **Methodology** section.

A. Homogeneous Ensemble

As per the **Methodology** section, a grid search was implemented to find the best hyper-parameter values for the DT, which then, would be used as hyper-parameter values in the RF. The following list is the best hyper-parameter values for the DT:

- *criterion* = ‘entropy’.
- *max_depth* = 25.
- *min_samples_leaf* = 1.
- *min_samples_split* = 5.

These values were used in the RF parameter values. Table 1 shows the results using the *RandomForestClassifier* from *scikit-learn*. A 10-fold cross validation was implemented to acquire more accurate results. The only parameter that was varied was the number of estimators in the RF algorithm. Furthermore, the tables shows the mean for the accuracy, weighted $F1$ and MCC scores, with their respective standard deviations and also the OOB score.

TABLE I
Performance metrics of RF for varying number of estimators.

n_estimators	Accuracy	$F1$ weighted	MCC
100	0.9508 (±0.0239)	0.9507 (±0.0238)	0.8953 (±0.0509)
200	0.9579 (±0.0178)	0.9577 (±0.0178)	0.9100 (±0.0388)
300	0.9526 (±0.0232)	0.9526 (±0.0230)	0.8993 (±0.0495)
400	0.9561 (±0.0277)	0.9560 (±0.0276)	0.9067 (±0.0592)
500	0.9561 (±0.0266)	0.9560 (±0.0265)	0.9066 (±0.0568)
600	0.9526 (±0.0251)	0.9525 (±0.0250)	0.8990 (±0.0537)
700	0.9579 (±0.0290)	0.9578 (±0.0289)	0.9104 (±0.0619)
800	0.9579 (±0.0268)	0.9578 (±0.0267)	0.9105 (±0.0571)
900	0.9579 (±0.0279)	0.9578 (±0.0278)	0.9103 (±0.0596)
1000	0.9596 (±0.0269)	0.9596 (±0.0268)	0.9140 (±0.0573)

TABLE II
OOB score of RF for varying number of estimators.

n_estimators	OOB score
100	0.9543
200	0.9543
300	0.9596
400	0.9613
500	0.9578
600	0.9631
700	0.9596
800	0.9578
900	0.9578
1000	0.9613

From Table 1 and 2, clearly 1000 estimators performs the best out of all the different amounts of estimators. However, any value would produce sufficient results and for any number of estimators tested, would produce a model which performs well. The following figure is the confusion matrix for a RF using 1000 estimators. Using *train_test_split()*, the Rf is trained and tested on the breast cancer dataset.

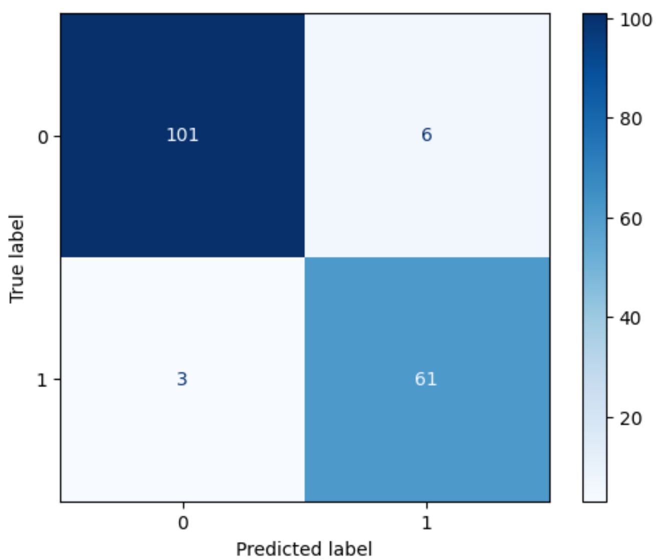

Fig. 5. Confusion matrix of RF.

The model demonstrates high overall accuracy and strong class separation, with only a few misclassification. It shows shows slightly more false positives than false negatives, meaning it occasionally predicts malignant for benign cases. Figure 6 is the ROC curve using the best hyper-parameter values found in the previous table:

The ROC curve shows that the RF trained on the original dataset performs exceptionally well in distinguishing between malignant and benign cases. The RF achieves excellent discriminative ability on the original dataset, showing strong generalisation and minimal misclassification, which indicates reliable predictive performance.

The previous results for the homogeneous ensemble were based on the original, skewed class distribution. The following results utilise a balanced distribution to accomplish two objectives: first, to determine if balancing alters the RF performance, and second, to enable a direct comparison with the heterogeneous ensembles.

For the balanced distribution dataset, a grid search was performed again to see if the best parameter values would be altered. The following list are the best hyper-parameter values from grid search for the DT:

- *criterion* = 'gini'.
- *max_depth* = 10.
- *min_samples_leaf* = 4.
- *min_samples_split* = 15.

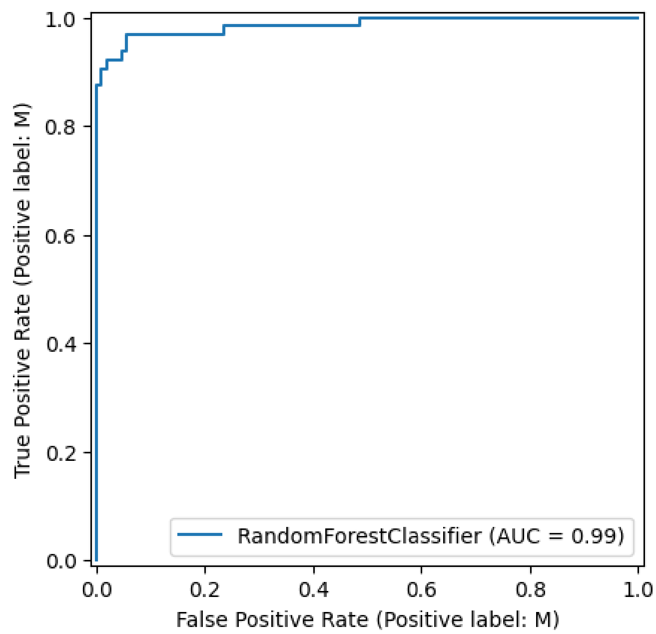

Fig. 6. ROC cruve for RF.

Clearly, there is a difference when comparing these values to the values obtained from the grid search on the original dataset.

Tables 3 and 4 shows the same results as Tables 1 and 2 but now for a balanced class distribution:

TABLE III
PERFORMANCE METRICS OF RF FOR VARYING NUMBER OF ESTIMATORS (FIXED CLASS DISTRIBUTION).

n_estimators	Accuracy	$F1$ weighted	MCC
100	0.9566 (±0.0239)	0.9566 (±0.0239)	0.9144 (±0.0465)
200	0.9566 (±0.0230)	0.9566 (±0.0230)	0.9143 (±0.0449)
300	0.9608 (±0.0241)	0.9608 (±0.0241)	0.9226 (±0.0471)
400	0.9608 (±0.0241)	0.9608 (±0.0241)	0.9226 (±0.0471)
500	0.9622 (±0.0198)	0.9622 (±0.0198)	0.9251 (±0.0390)
600	0.9622 (±0.0198)	0.9622 (±0.0198)	0.9251 (±0.0390)
700	0.9636 (±0.0209)	0.9636 (±0.0209)	0.9279 (±0.0413)
800	0.9622 (±0.0198)	0.9622 (±0.0198)	0.9251 (±0.0390)
900	0.9636 (±0.0209)	0.9636 (±0.0209)	0.9279 (±0.0413)
1000	0.9622 (±0.0198)	0.9622 (±0.0198)	0.9251 (±0.0390)

TABLE IV
OOB SCORE OF RF FOR VARYING NUMBER OF ESTIMATORS.

n_estimators	OOB score
100	0.9608
200	0.9608
300	0.9622
400	0.9608
500	0.9608
600	0.9608
700	0.9608
800	0.9594
900	0.9608
1000	0.9608

Comparing Tables 3 and 4 with Tables 1 and 2, the RF performs worse on the balanced distribution. This could be attributed to the fact that RF is robust to skewed class distributions. When balancing the distributions, *SMOTE* produces synthetic instances, which could be influencing the performance of the RF negatively.

The following two figures are the confusion matrix and the ROC curve, respectively, for the best possible hyper-parameter values for the RF using the balanced dataset:

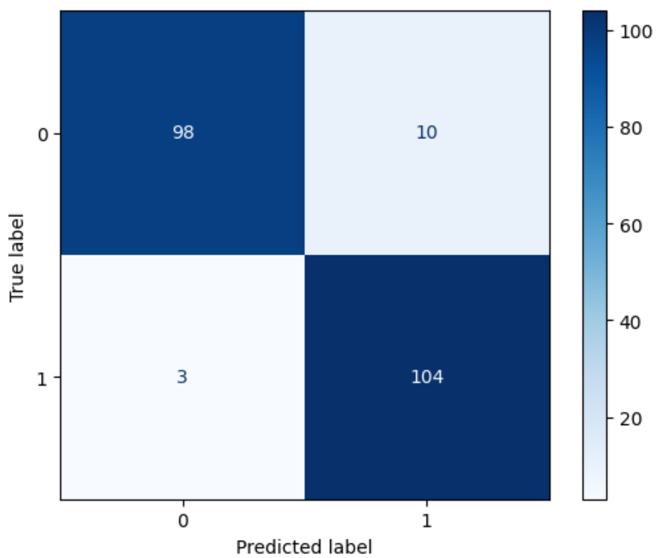

Fig. 7. Confusion matrix for RF (fixed class distribution).

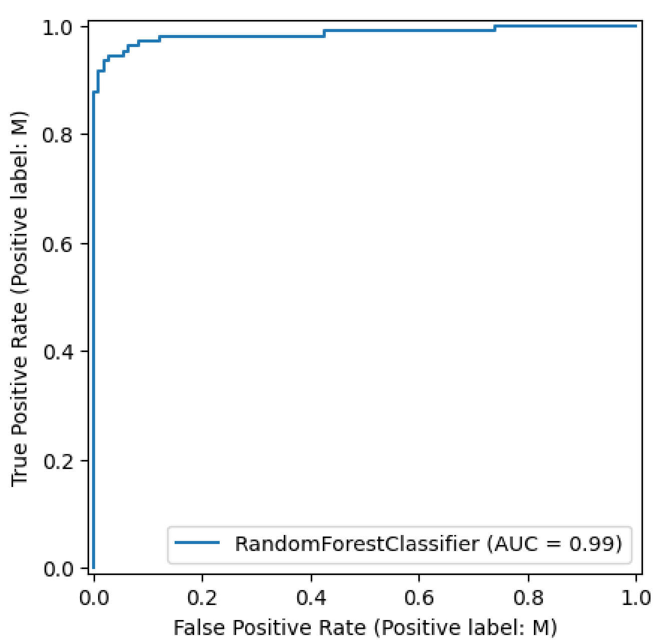

Fig. 8. ROC curve for RF (Fixed class distribution).

From Figure 7, the confusion matrix shows that the model misclassifies more often than with the original dataset. However, from Figure 8 the models ability to distinguish between classes hasn't changed.

For the homogeneous ensemble, using the original breast cancer dataset proved to yield better performance (slightly better).

B. Heterogeneous Ensemble

As per the **Methodology** section, a grid search was used in order to find the best hyper-parameter values for each algorithm within the heterogeneous ensemble. The following list displays the best hyper-parameter values for each algorithm while using grid search for the original breast cancer dataset:

LR:
- *C* = 10.
- *l1_ratio* = 0.0.
- *penalty* = l2.
- *solver* = 'lbfgs'.

kNN:
- *n_neighbors* = 9.
- *p* = 1.
- *weights* = 'uniform'.

DT:
- *criterion* = 'entropy'.
- *max_depth* = 40.
- *min_samples_leaf* = 4.
- *min_samples_split* = 2

GB:
- *learning_rate* = 0.05.
- *max_depth* = 4.
- *n_estimators* = 300.
- *subsample* = 0.8

SVM:
- *C* = 100.
- *gamma* = 'scale'.
- *kernel* = 'rbf'.

The following table shows the best cross validation accuracy score by displaying the *best_score_* from using the *GridSearchCV* function. The results will help us determine the weights for the algorithms to be used in the *VotingClassifier* heterogeneous ensemble:

TABLE V
ACCURACY SCORE FROM GRID SEARCH RESULTS USING 5-FOLD CROSS VALIDATION.

Algorithm	Accuracy score
LR	0.9471
kNN	0.9471
DT	0.9423
GB	0.9623
SVM	0.9346

Based on the results in Table 5, the weight parameter for the *VotingClassifier* was set to [1, 1, 1, 2, 1]. This weighting scheme assigns greater influence to the GB algorithm, reflecting its superior individual performance relative to the other base models.

Once the hyper-parameters have been tuned individually for each algorithm, a heterogeneous ensemble was built using the best hyper-parameter values and a 10-fold cross validation was utilised in order to lower the variance in the results but ultimately, provide more reliable results. Table 6 shows the results:

TABLE VI
PERFORMANCE RESULTS FROM THE HETEROGENEOUS ENSEMBLE.

Accuracy	$F1$ weighted	MCC
0.9561 (±0.0378)	0.9557 (±0.0384)	0.9058 (±0.0821)

From Table 6, the heterogeneous ensemble performs well with the original breast cancer dataset. However, clearly the homogeneous ensemble is superior in this case, when compared to the heterogeneous ensemble. The homogeneous ensemble was able to achieve a higher accuracy, $F1$ weighted and MCC score than the heterogeneous ensemble. Nonetheless, both the homogeneous and heterogeneous ensembles performs well for the original breast cancer dataset.

Figures 9 and 10 shows the confusion matrix and ROC curve for the heterogeneous ensemble when applied to the original breast cancer dataset, respectively:

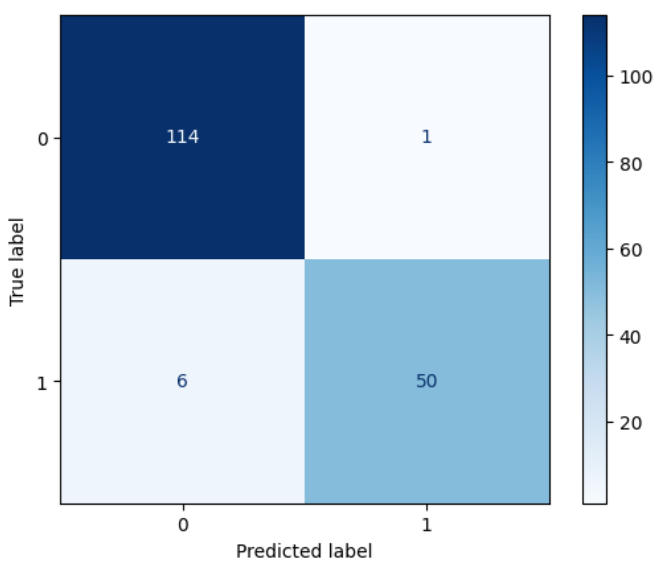

Fig. 9. Confusion matrix of heterogeneous ensemble.

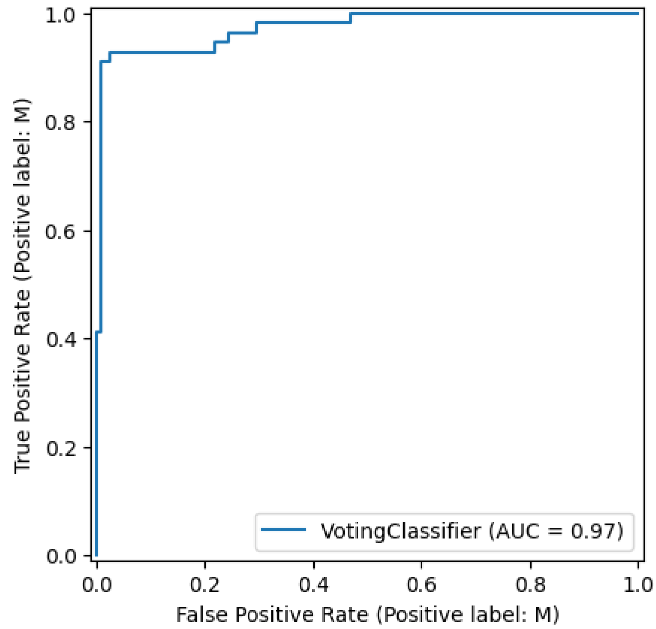

Fig. 10. ROC curve of heterogeneous ensemble.

Figure 9 shows that the heterogeneous misclassifies more often than the RF. This makes sense, considering earlier performance metrics for the heterogeneous ensembles were lower than the homogeneous ensemble (RF). Figure 10 shows that the heterogeneous ensemble This means ranks malignant samples higher than benign ones 97% of the time, indicating a high level of separability between the two classes. However, when compared to the RF ROC curve (refer to Figure 6), the RF is slightly better at detecting class boundaries.

The previous results for the the heterogeneous ensemble were achieved using the original breast cancer dataset. For the final part of the **Research Results** section, the skewed class distribution was balanced using *SMOTE* and new synthetic instances were made to balance the class distribution. Again, a grid search with 5-fold cross validation was performed to see if addressing the skewed class distributions affected the results from a grid search. The following lists are the results from grid searches applied to all machine learning algorithms within the heterogeneous ensemble individually:

LR:

- *C* = 10.
- *l1_ratio* = 0.0.
- *penalty* = l2.
- *solver* = 'lbfgs'.

kNN:

- *n_neighbors* = 5.
- *p* = 3.
- *weights* = 'distance'.

DT:

- *criterion* = 'gini'.

- *max_depth* = 20.
- *min_samples_leaf* = 4.
- *min_samples_split* = 15.

GB:

- *learning_rate* = 0.1.
- *max_depth* = 2.
- *n_estimators* = 100.
- *subsample* = 0.8

SVM:

- *C* = 1.
- *gamma* = 0.001.
- *kernel* = 'rbf'.

Indeed, balancing the class distribution did affect the results from the grid search. Adjusting these hyper-parameter values within the *VotingClassifier*, the following table illustrates our results using a 10-fold cross validation on the balanced class distribution dataset:

TABLE VII
PERFORMANCE RESULTS FROM THE HETEROGENEOUS ENSEMBLE (FIXED CLASS DISTRIBUTION).

Accuracy	$F1$ **weighted**	**MCC**
0.9525 (±0.0360)	0.9522 (±0.0365)	0.8991 (±0.0778)

Interestingly, the performance seemed to get worse when balancing the class distribution. However, the standard deviations in the scores have lowered. This could be a results of the new synthetic results produced when using *SMOTE*.

Figures 11 and 12 show the confusion matrix and the ROC curve for the heterogeneous ensemble when applied to the balanced class distribution dataset:

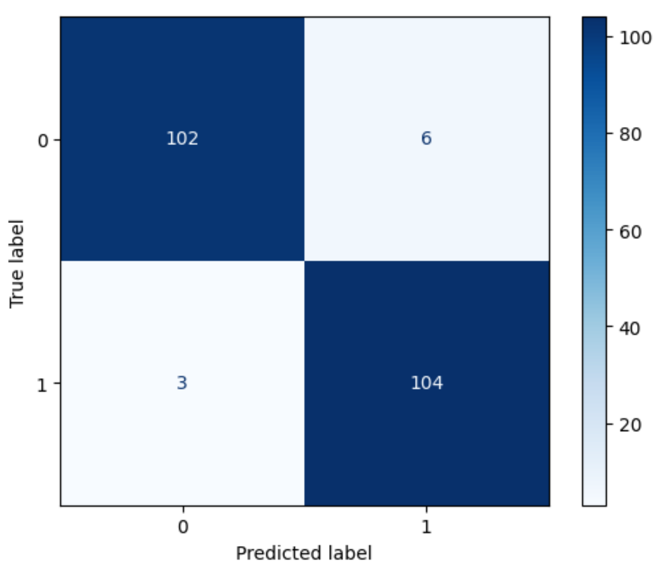

Fig. 11. Confusion matrix for the heterogeneous ensemble when using balanced class distributions.

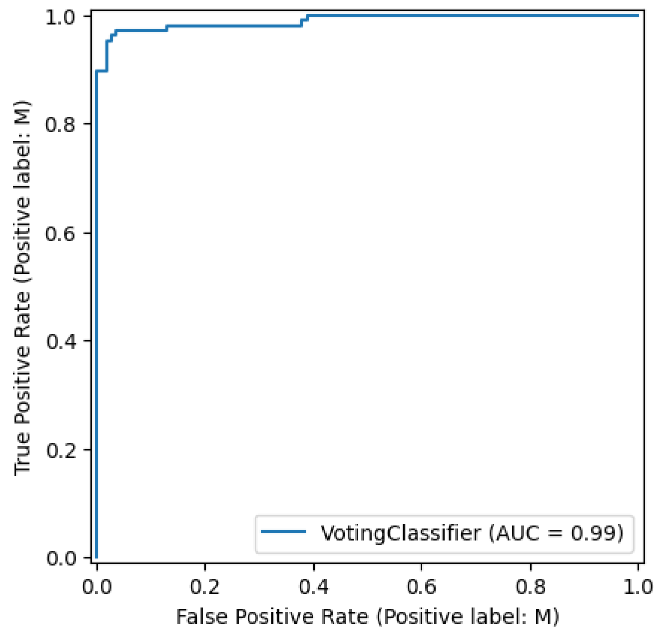

Fig. 12. Roc curve for heterogeneous ensemble using balanced class distributions.

From Figure 11 fixing the skewed class distribution did not necessarily improve performance nor does it perform better then the RF. However, from Figure 12, the ROC curve has an AUC of 0.99, reflecting almost perfect separability between malignant and benign classes, which is an improvement from Figure 10.

VI. CONCLUSION

This study evaluated a homogeneous ensemble based on a Random Forest (RF) and a heterogeneous ensemble formed with Logistic Regression (LR), k-Nearest Neighbours (kNN), Decision Tree (DT), Gradient Boosting (GB) and Support Vector Machine (SVM) combined by soft voting. The analysis used cross-validation, hyper-parameter tuning using grid search, and multiple metrics, including accuracy, weighted F1, and Matthews Correlation Coefficient (MCC) scores. Results showed that the RF achieved the strongest overall performance on the original breast cancer dataset. The heterogeneous ensemble performed competitively, but underperformed the RF across the primary metrics.

The comparative findings supported two observations. First, the RF outperformed the heterogeneous ensembles only slightly. As this dataset is not very large or complicated, both homogeneous and heterogeneous ensembles performed very similar. Secondly, when balancing the class distributions, it did not increase performance and in some aspects decreased the performance of the ensembles.

The work demonstrated that a well tuned RF provided a strong and robust baseline for this classification problem and that, under the tested configurations, a heterogeneous soft voting ensemble did not offer a advantage. Future work could investigate different algorithms for both the homogeneous ensemble and heterogeneous ensemble. Additionally, a more complex dataset could be utilised in order to fully compare these two approaches . Overall, the results indicated that the homogeneous ensemble (RF) is preferred approach for this dataset however, the heterogeneous ensemble is also a good approach.

Git Repository link:
https://github.com/CJW116/ML741-Assignment-4

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